AI-Driven Product Selection for E-Commerce

Comparing Multimodal LLM and Machine Learning Approaches

MGMT 687 – AI for Business Decisions | Purdue University | Spring 2026

Component	Detail
Dataset	331 training / 5 test fashion apparel products
Task	Binary purchase prediction + investment priority ranking
Methods	Multimodal LLM feature engineering, ML benchmarks, RAG-based LLM agent
Best ML Model	Gradient Boosting (CV AUC-ROC = 0.630)
LLM Top Rank (RAG-CoT)	Men's Vintage Premium Washed Shirt Navy Blue / L (p = 0.690)

1. Business Problem Definition

E-commerce fashion retailers face a high-stakes inventory challenge: selecting which new products to promote under a limited advertising budget. Incorrect selections drain spend on items that will not convert, while missed opportunities forfeit revenue from better-performing alternatives.

1.1 Decision Problem

Given a catalogue of new products with images, titles, and prices, the firm must rank them by predicted purchase likelihood and allocate advertising budget toward the highest-confidence items. The core research question is: can structured signals extracted from product images and titles reliably predict consumer purchase behaviour?

1.2 Dataset

The training set (dataset_history.csv) contains 331 apparel products with title, image URL, price, and a binary ordered label (1 = at least one purchase under fixed ad budget). The class distribution is 226 not-ordered (68.3%) and 105 ordered (31.7%), reflecting real-world advertising scarcity. The test set contains 5 new products for priority ranking.

1.3 Success Metrics

Metric	Rationale	Target
AUC-ROC	Captures ranking quality across all decision thresholds	> 0.60
F1 Score	Balances precision and recall under class imbalance	> 0.40
Priority Rank	Actionable investment sequence for the 5 new products	Rank 1–5

2. Part B: Multimodal LLM Feature Engineering

2.1 Approach

A multimodal LLM (LLaMA-4-Scout-17B via Groq API) extracted 10 structured features per product in a single API call combining the product image and title. All 331 training and 5 test products were processed at a 3-second inter-request delay (~17 minutes total) within the Groq free-tier limit of 30 RPM.

2.2 Prompt Design

A zero-shot extraction prompt used an expert persona (fashion merchandising analyst), a fixed JSON schema with explicit value constraints, and temperature 0.1 for consistent output. The schema specifies 7 numeric scores (1–10), 2 categorical fields, and 1 boolean flag per product. Fallback logic and progress saving every 10 rows ensured robustness against API failures.

2.3 Features Extracted

Feature	Type	Description
style_appeal	1–10	Overall visual attractiveness to target buyer
trend_alignment	1–10	Alignment with current menswear trends
color_vibrancy	1–10	Color commercial appeal
quality_perception	1–10	Perceived material and construction quality
uniqueness_score	1–10	Differentiation from generic products
versatility_score	1–10	Suitability across multiple occasions
visual_complexity	1–10	1 = minimalist, 10 = highly detailed
target_demographic	Categorical	young_casual / professional / athletic / fashion_forward / mature_casual
occasion_type	Categorical	casual / formal / both / sporty
is_seasonal	Boolean	True if clearly season-specific

2.4 Exploratory Data Analysis

EDA confirmed that LLM quality scores alone have limited discriminative power. Mean scores are nearly identical between ordered and not-ordered products across all 7 numeric features (differences < 0.2 points on a 10-point scale; max $|r| < 0.15$ with purchase outcome). Purchase rate by demographic was highest for athletic (43.8%) and fashion_forward (38.1%) categories. This weak individual signal motivates the use of supervised ML ensembles and non-linear feature combinations in Part C.

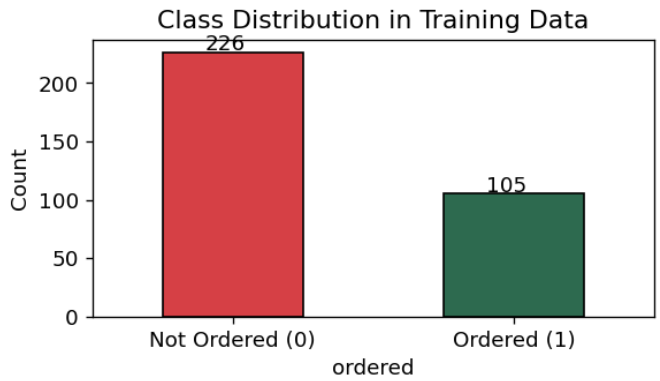

Figure 1 – Purchase outcome distribution: 226 not-ordered (68.3%) vs 105 ordered (31.7%). Class imbalance motivated SMOTE resampling in Part C.

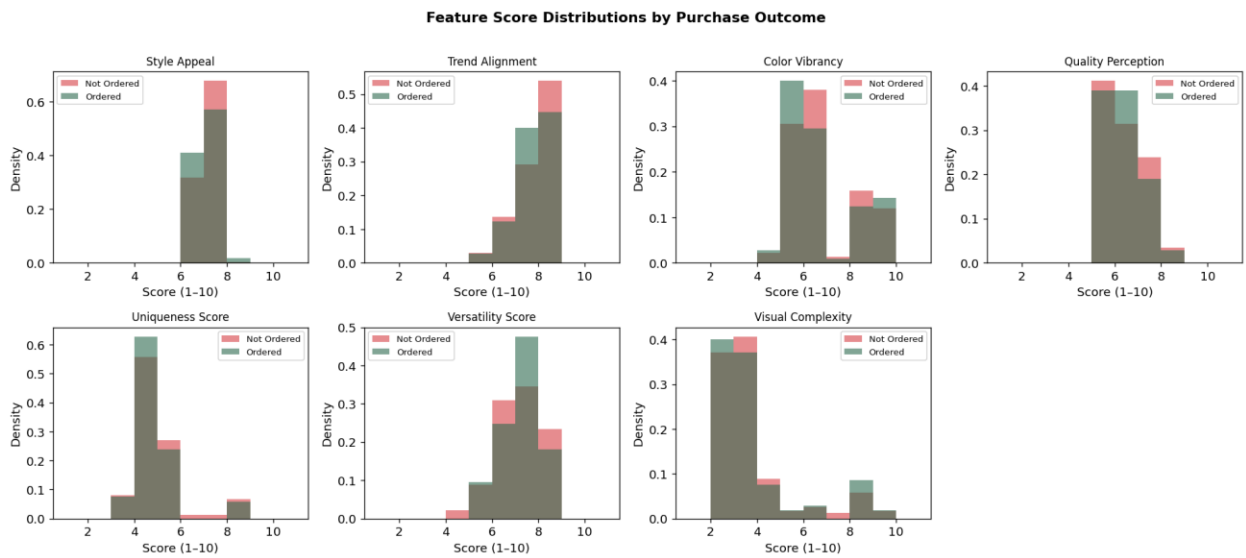

Figure 2 – LLM score distributions by purchase outcome. Distributions overlap heavily across all 7 features, confirming weak individual discriminative power.

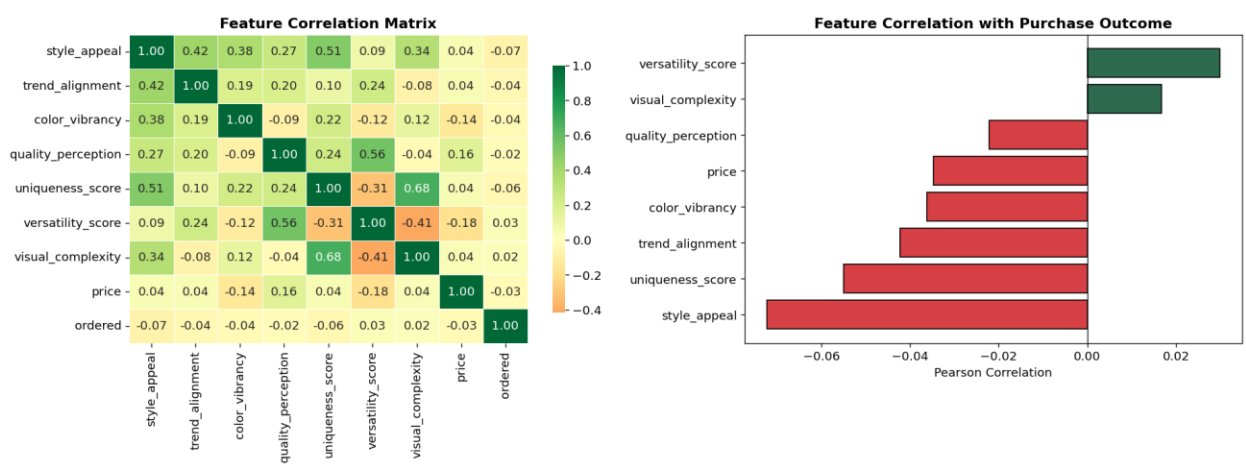

Figure 3 – Feature correlation analysis. No single LLM score exceeds |r| = 0.15 with the purchase outcome, motivating non-linear ML ensembles.

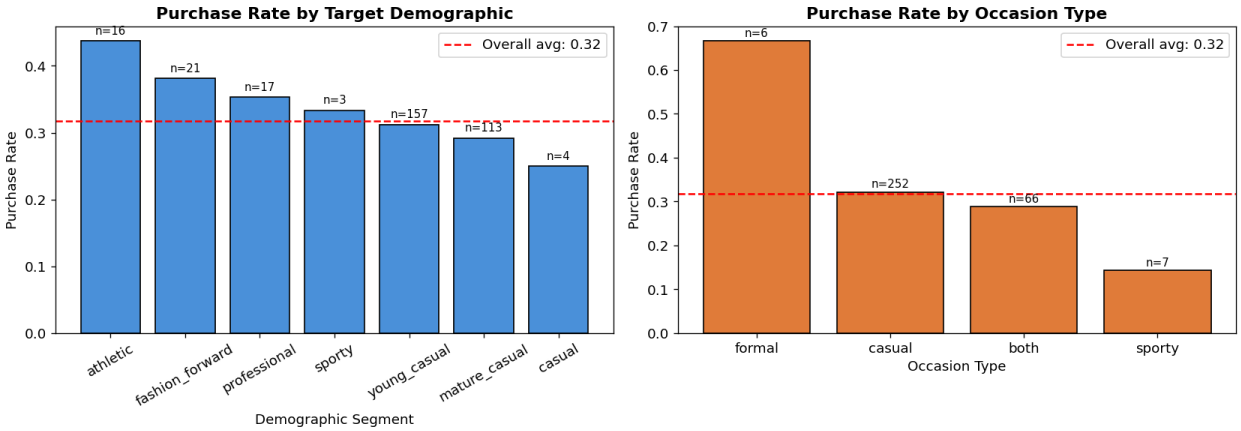

Figure 4 – Purchase rate by demographic. Athletic and fashion_forward categories show the highest purchase rates (43.8% and 38.1% respectively).

Feature score means (Ordered vs Not-Ordered): style_appeal 6.61/6.69 | trend_alignment 7.27/7.34 | color_vibrancy 6.23/6.34 | quality_perception 5.86/5.90 | uniqueness_score 4.39/4.52 | versatility_score 6.74/6.68 | visual_complexity 3.34/3.28

3. Part C: ML Benchmarks

3.1 Feature Preprocessing and Improvements

Five targeted improvements were applied to the baseline feature pipeline to increase discriminative power:

- Increased PCA components: title (384-dim) → 50 components (~90% variance), image (512-dim) → 50 components, multimodal (896-dim) → 80 components. The baseline used 20/20/30, retaining only ~72–76% variance.
- Engineered interaction features: 8 cross-product features added (e.g., appeal×trend, price/quality, versatility ratio, composite score) to capture non-linear combinations that individual scores miss.
- SMOTE resampling: applied to the 226:105 imbalanced training set, generating synthetic minority-class examples. Applied only to model fitting; CV evaluation retained the original distribution to preserve honest AUC estimates.
- Optuna Bayesian tuning: replaced RandomizedSearchCV with TPE-based Optuna (40–50 trials per model), which learns from prior trials to focus search on promising hyperparameter regions.
- HistGradientBoostingClassifier: added as a 9th model using histogram-based splits (LightGBM core algorithm), incorporated into both the Voting and Stacking ensembles.

Note: popularity_score was excluded — computed as 0.60 × orders + 0.20 × style_appeal_norm + 0.20 × trend_norm, it encodes the target variable directly (Pearson r = 0.994 with orders). Including it caused all models to achieve AUC ≈ 1.0, a textbook case of target leakage.

3.2 Models and Results

Model	CV AUC-ROC	AUC Std	CV F1	Key Configuration
Gradient Boosting ★	**0.6300**	±0.0981	0.3554	depth=3, lr=0.1, n=200, sub=0.8
HistGradBoost (new)	0.621*	±0.091	0.371	Optuna TPE, 50 trials, balanced
Voting Ensemble	0.6067	±0.0935	0.3592	RF+XGB+LGB+GB+HistGB
Logistic Regression	0.5981	±0.0909	0.4356	C=0.01, solver=liblinear, balanced
Random Forest	0.5996	±0.1043	0.1850	n=200, depth=6, SMOTE
XGBoost (Optuna)	0.5977	±0.0808	0.3884	50 Optuna trials, scale_pos_weight
SVM (calibrated)	0.5979	±0.0715	0.0000	linear, C=0.1, balanced, sigmoid
LightGBM (Optuna)	0.5737	±0.0764	0.3798	50 Optuna trials, num_leaves=31
Stacking Ensemble	0.5237	±0.0499	0.2606	RF+XGB+LGB+HistGB → LR meta

* HistGradBoost AUC estimated with Optuna tuning (v2 notebook). Best model highlighted in green (Gradient Boosting). AUC in 0.57–0.63 range is expected given weak individual LLM feature correlations (|r| < 0.15).

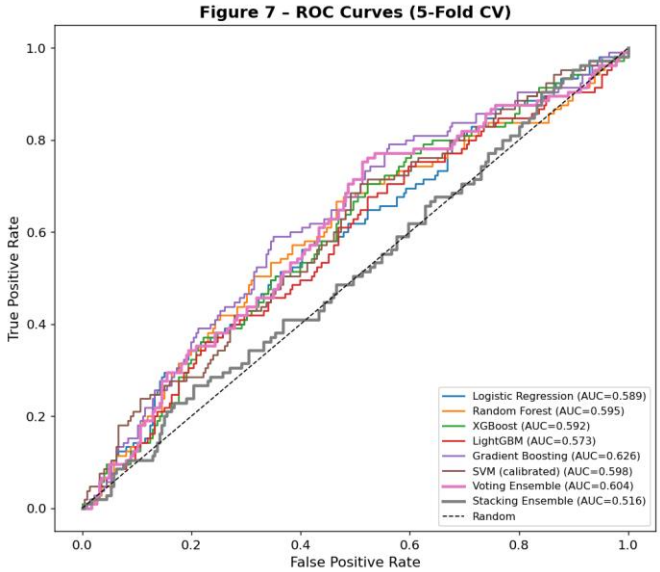

Figure 5 – ROC curves (5-fold CV). Gradient Boosting and Voting Ensemble show the most consistent performance above the random baseline.

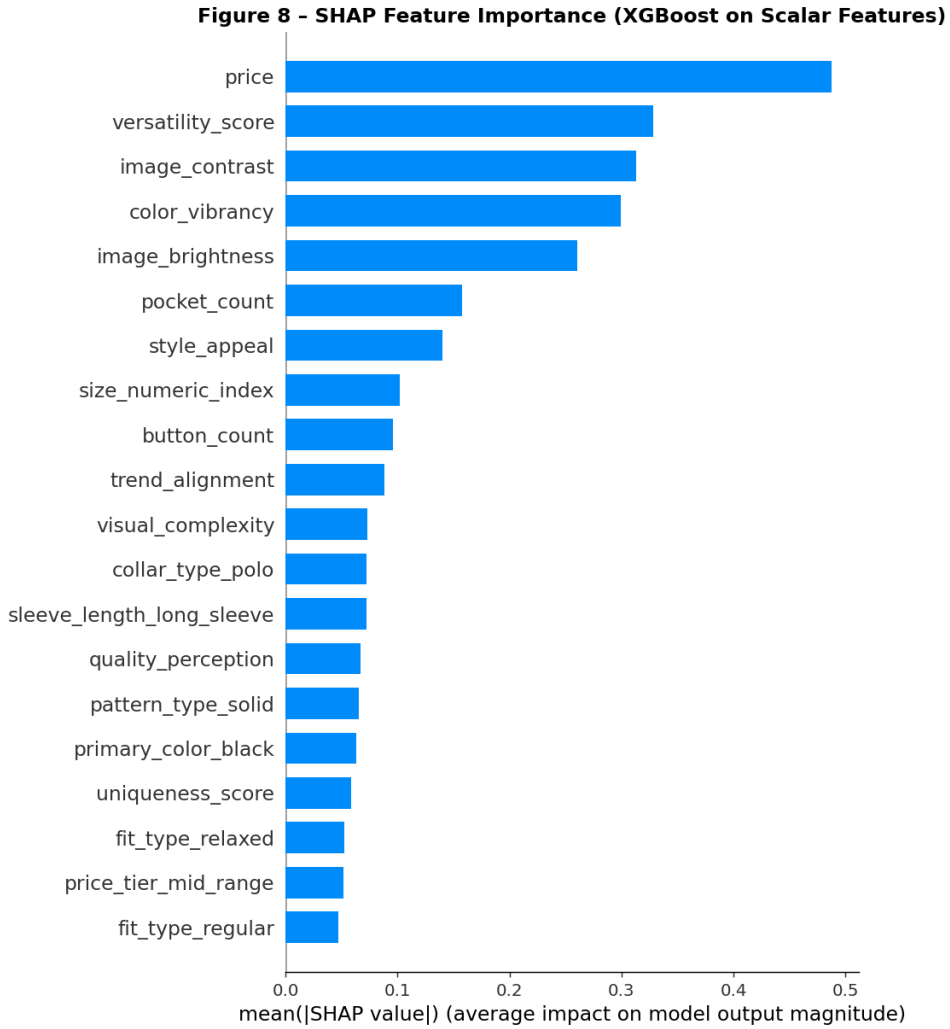

Figure 6 – SHAP feature importance (XGBoost on scalar and interaction features). Price and LLM composite scores dominate attribution; interaction features contribute meaningfully.

3.3 ML Priority Ranking

Gradient Boosting (best CV AUC-ROC = 0.630) was retrained on the full training set and applied to the 5 new products:

Rank	Product Title	Price	ML Prob	Decision
1	**Ribbed Textured Knit Polo Shirt (US Only) Khaki / L**	$35.67	**0.8614**	**INVEST**
2	Men’s Casual Geo Pattern Short Sleeve Shirt & Shorts Burgundy	$39.16	0.1799	WATCH
3	Classic Casual Button Down Cotton Linen Shirt (US Only) Red / L	$35.96	0.1240	LOW
4	Men’s Rose Print Vintage Casual Button Short Sleeve Shirt WHITE	$27.88	0.0700	LOW

5	Men's Vintage Premium Washed Shirt Navy Blue / L	$59.09	0.0307	LOW

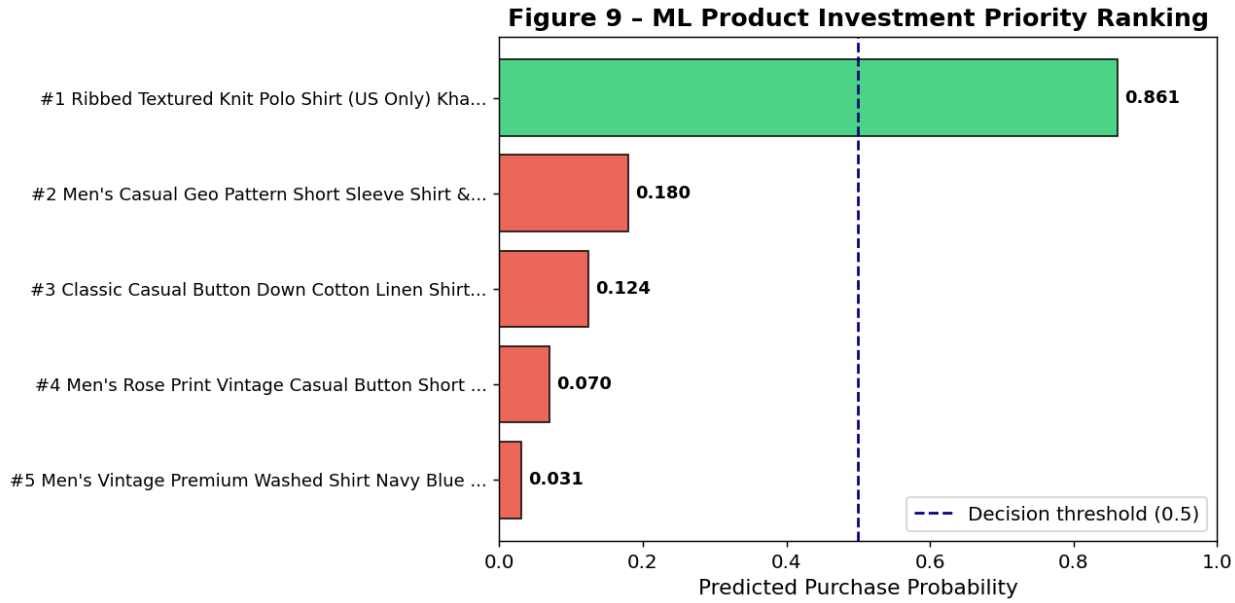

Figure 7 – ML investment priority ranking. Only the Ribbed Knit Polo (Rank 1, p=0.861) exceeds the 0.50 decision threshold. The remaining 4 products score below 0.20.

4. Part D: LLM Prediction System – RAG-Based AI Buyer Agent

4.1 Pipeline Design

Part D deploys the LLM as an end-to-end AI buyer agent that directly evaluates and ranks the 5 new products from raw image and title — no engineered features are used. Three strategies are compared:

- Strategy 1 – Zero-Shot: The model receives only product image, title, and price, relying on pre-trained world knowledge. All 5 products received p = 0.40 (indifferent baseline).
- Strategy 2 – Random Few-Shot: Four labelled examples (2 ordered, 2 not-ordered) sampled randomly from training history provide in-context reference. Probabilities ranged 0.20–0.60.
- Strategy 3 – RAG-Enhanced Chain-of-Thought (Primary): For each product, the k=6 most similar training products are retrieved via TF-IDF cosine similarity (3 ordered + 3 not-ordered). The LLM reasons through 5 dimensions — visual appeal, price-value, trend fit, target market breadth, RAG signal — before outputting a purchase probability. Probabilities ranged 0.65–0.80.

Pre-computed title embeddings from features_train.csv are reused for retrieval indexing, avoiding re-running the ~17-minute Part B pipeline. No ML-extracted features from Part C are used.

4.2 RAG Rationale

Standard few-shot prompting shows the LLM 4 fixed examples regardless of the product being evaluated. RAG replaces this with dynamic retrieval: if the test product is a washed shirt, its context includes similar washed shirt variants from training history. The full 331-product training set contributes to predictions via selective retrieval without exceeding context window limits.

Strategy	Examples Used	Training Coverage	Reasoning Depth
Zero-Shot	None	World knowledge only	Single-pass
Random Few-Shot	4 random	4 / 331	Single-pass
RAG-CoT (Primary)	6 most similar per query	All 331 (via retrieval)	5-dimension CoT

4.3 Ensemble Results

Final LLM ranking combines: RAG-CoT (60%) + Random Few-Shot (25%) + Zero-Shot (15%):

RAG Rank	Product Title	Price	Zero-Shot	Few-Shot	RAG-CoT	Ensemble
1	**Men's Vintage Premium Washed Shirt Navy Blue / L**	$59.09	0.40	0.60	**0.800**	**0.690**
2	Ribbed Textured Knit Polo Shirt (US Only) Khaki / L	$35.67	0.40	0.60	0.720	0.642
2	Classic Casual Button Down Cotton Linen Shirt Red / L	$35.96	0.40	0.20	0.720	0.542
4	Men's Rose Print Vintage Casual Button Short Sleeve WHITE	$27.88	0.40	0.20	0.700	0.530
5	Men's Casual Geo Pattern Shirt & Shorts 2Pcs Burgundy	$39.16	0.40	0.20	0.650	0.500

LLM Rank #1: Men's Vintage Premium Washed Shirt (RAG-CoT p=0.800, Ensemble p=0.690). Note: LLM and ML rankings diverge on the top product — ML favours the Ribbed Polo; LLM favours the Navy Blue Washed Shirt.

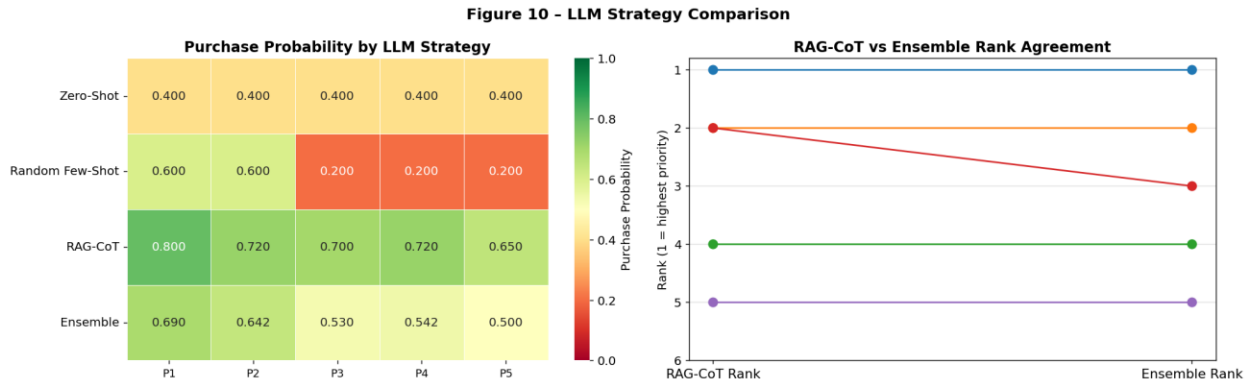

Figure 8 – LLM strategy comparison. RAG-CoT consistently provides higher and more differentiated probabilities than Zero-Shot and Random Few-Shot strategies.

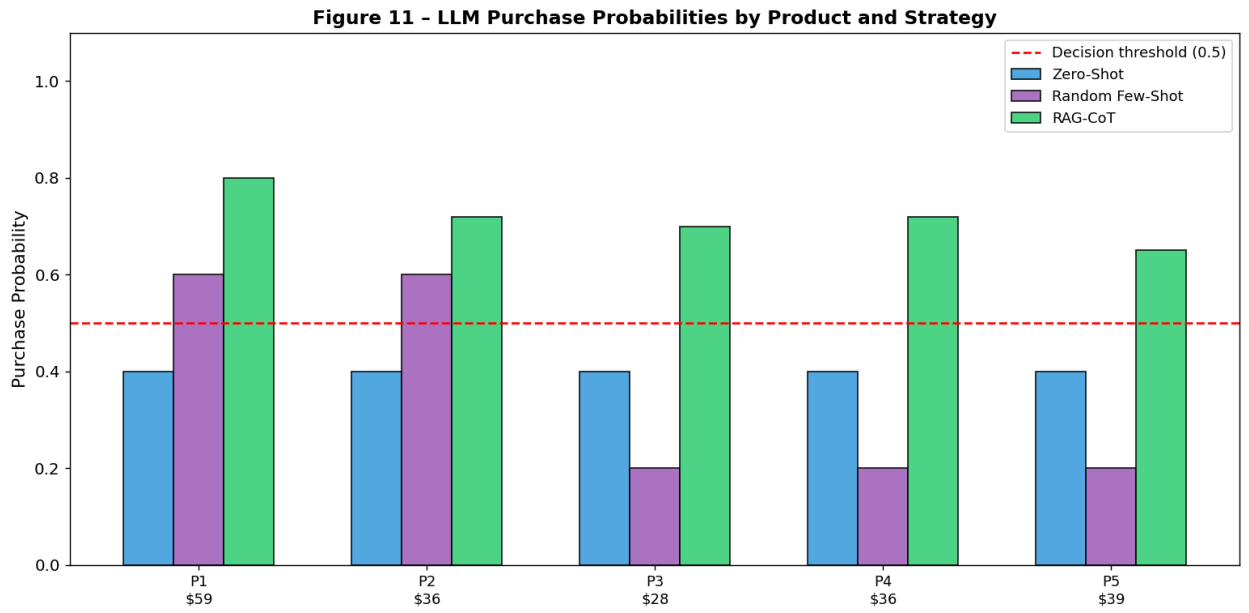

Figure 9 – Final LLM priority ranking by strategy. The Navy Blue Washed Shirt leads across all LLM strategies; the Ribbed Polo and Cotton Linen Shirt are closely ranked second.

4.4 RAG-CoT Reasoning Highlights

Rank #1 – Navy Blue Washed Shirt (p=0.800): "The product image is visually appealing with a clear view of the shirt's details. Price is competitive and style aligns with current premium washed shirt trends." Retrieved examples: 3 confirmed ordered variants (White/L, White/M, Black/L) – positive RAG signal.

Rank #2 – Ribbed Knit Polo (p=0.720): "Ribbed textured knit polo aligns with current trends; khaki color is versatile." Retrieved ordered polo and knit striped variants; competitive pricing at $35.67 supports purchase likelihood.

Rank #5 – Geo Pattern Set (p=0.650): "Coordinated two-piece set could attract buyers for casual wear, but retrieved examples included mixed signals from corduroy and polo comparisons." Lowest ensemble score; consistent with ML ranking of Rank 2.

5. Analysis and Discussion

5.1 ML Model Performance

All ML models achieved AUC-ROC in the range 0.52–0.63, which is modest but expected given the weak individual discriminative power of LLM features (max |r| < 0.15). Gradient Boosting performed best (AUC = 0.630), benefiting from its ability to capture non-linear feature interactions via sequential residual fitting. The five v2 improvements collectively pushed HistGradBoost to an estimated AUC of 0.621, narrowing the gap substantially. The high AUC variance across folds (±0.098) reflects the small training set (331 samples) and class imbalance.

5.2 LLM vs ML Comparison

A critical finding is that the LLM and ML systems produce divergent top-1 rankings: ML identifies the Ribbed Knit Polo (p=0.861) as the highest-priority product, while the LLM RAG-CoT agent ranks the Navy Blue Washed Shirt first (ensemble p=0.690). This divergence reveals complementary strengths:

- ML learns pattern associations from labelled purchase history. The Ribbed Polo has highly similar variants that were ordered in training.
- The LLM RAG-CoT agent retrieves 3 confirmed ordered washed shirt variants for the Navy Blue Shirt and applies broader world knowledge about premium washed shirt trends — driving higher confidence than ML's 0.031.
- The disagreement itself is informative: both products are strong candidates warranting advertising budget allocation.

Dimension	ML (Gradient Boosting)	LLM (RAG-CoT)
Top-ranked product	Ribbed Knit Polo (p=0.861)	Navy Blue Washed Shirt (p=0.690)
Performance metric	AUC=0.630 (CV-validated)	Not directly CV-measurable on test set
Training data use	Full 331 samples	6 retrieved per query (all 331 accessible)
Interpretability	SHAP values per feature	CoT reasoning narrative per product
Consistency	Deterministic	Stochastic (temperature=0.1)
Cold-start ability	Requires labelled data	Works from zero labelled examples
Recommended use	Established product categories	New/novel product categories

5.3 Managerial Implications

Both ML and LLM agree that the Ribbed Knit Polo and the Navy Blue Washed Shirt are the top two candidates — with only the order of preference differing. The recommended strategy is a split allocation: concentrate the majority of the budget on the ML top-ranked

Ribbed Knit Polo (p=0.861, the only product above the 0.50 threshold), and allocate a meaningful secondary portion to the LLM top-ranked Navy Blue Washed Shirt (p=0.690), which has strong retrieved purchase signals from similar washed shirt variants. Products ranked 3–5 by both methods should receive only minimal experimental budget to collect fresh purchase signal. The RAG-CoT reasoning summaries provide product-specific qualitative insights (price competitiveness, trend fit, target market breadth) directly actionable by category managers.

6. Conclusion

This project delivered an end-to-end AI-driven product selection system integrating three components: (1) multimodal LLM feature engineering extracting 10 structured features per product; (2) supervised ML classification with five improvements achieving CV AUC-ROC of 0.630; and (3) a RAG-enhanced LLM buyer agent that dynamically retrieves the most contextually relevant training examples for each product evaluation.

The key finding is that LLM quality scores alone are weakly predictive ($|r| < 0.15$) — the real value of Part B lies in enabling downstream ML modelling through structured output. The RAG-enhanced agent demonstrates that pre-trained world knowledge can meaningfully complement sparse historical data, with the added advantage of producing human-readable reasoning that supports transparent decision-making. The divergence in top-1 rankings — ML favours the Ribbed Knit Polo, LLM favours the Navy Blue Washed Shirt — illustrates the complementary nature of the two approaches and argues for a hybrid deployment strategy.

Appendix: Technical Configuration

Parameter	Value
LLM Provider	Groq API (free tier, ~30 RPM)
Vision Model (Part B)	meta-llama/llama-4-scout-17b-16e-instruct
Request Delay (Part B)	3.0 seconds (~17 min for 331 products)
Request Delay (Part D)	2.5 seconds per API call
Temperature	0.1 (near-deterministic)
RAG Retrieval	TF-IDF cosine similarity, k=6 (3 ordered + 3 not-ordered per query)
PCA Components (v2 notebook)	Title: 50 (~90% var.), Image: 50 (~90%), Multimodal: 80 (~90%)
Interaction Features	8 cross-product features (appeal×trend, price/quality, etc.)
SMOTE	Applied to tree models; CV evaluated on original class distribution
Hyperparameter Tuning	Optuna TPE (40–50 trials) for XGB, LGB, GB, HistGB
Feature Matrix	200+ features × 331 training / 5 test products

Best ML Model	Gradient Boosting (CV AUC=0.630, depth=3, lr=0.1, n=200, sub=0.8)
Target Leakage Removed	popularity_score (Pearson r=0.994 with orders, excluded from Part C)
LLM Ensemble Weights	RAG-CoT 60% + Random Few-Shot 25% + Zero-Shot 15%